

Methods in Computational Linguistics

Correlation and Regression

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Today's slides are based on the materials provided by Eva Maria Vecchi.

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

What examples of statistics have we covered in the lecture so far?

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
 - ▶ identify **recurring trends and anomalies** within language usage
 - ▶ provides empirical evidence for linguistic theories
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
 - ▶ better understand language usage, word popularity
 - ▶ determine essential features of corpora, e.g., common vs. rare words or phrases
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
 - ▶ provides the tools to assess and compare the performance of different linguistic models (e.g., machine learning models for NLP)
 - ▶ accuracy, precision, recall, and F1 score are used to determine how well a model interprets or generates linguistic data
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
 - ▶ helps in explaining why a model behaves in certain ways by revealing correlations and dependencies between variables
 - ▶ e.g., how specific features of language influence model outcomes
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
 - ▶ test whether a specific language pattern/phenomenon is statistically significant
 - ▶ see if a proposed linguistic theory holds across diverse datasets
- ▶ Generalizability of Findings

Statistics

Statistics characterizes the results of previous experiments and uses them to **predict** the results of future implementations

- ▶ Quantifying Linguistic Patterns
- ▶ Frequency Analysis
- ▶ Model Performance Evaluation
- ▶ Interpretability and Explanation
- ▶ Hypothesis Testing
- ▶ Generalizability of Findings
 - ▶ make inferences that generalize beyond a specific dataset
 - ▶ ensures that conclusions drawn from linguistic data are not merely coincidental or overfitted to a particular sample

Correlation

Correlation

Correlation between two attributes is a measure of the **strength of association** between them

Correlation Coefficient	Interpretation
1	Attributes perfectly associated: if x grows, y grows as well
0	Attributes unrelated
-1	Attributes perfectly associated: if x grows, y shrinks

Linear correlation relationship overview

- Uses: **Data analysis and Evaluation**
 - Which **predictor** attributes is correlated with the **predicted** attribute?

Types of Correlation

- ▶ Pearson's r

- ▶ measures the **linear relationship** between two continuous variables (parametric)
- ▶ assumes that both variables are normally distributed and their relationship is linear

- ▶ Spearman's ρ (rho)

- ▶ measures a **rank correlation** as a monotonic relationship between two variables (non parametric)
- ▶ assesses whether the variables tend to move in the same direction (increasing or decreasing) but not necessarily at a constant rate

Pearson's r

Parametric correlation coefficient, linear relationship

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Where:

- ▶ x and y are the individual sample points (data points) for the two variables
- ▶ n is the number of data points
- ▶ $\sum x$ and $\sum y$ are the sums of the x -values and y -values
- ▶ $\sum x^2$ and $\sum y^2$ are the sums of the squares of the x -values and y -values
- ▶ $\sum xy$ is the sum of the product of corresponding x and y values

Example: Pearson's r

Calculate Pearson's r for variables x and y :

x	y
1	2
2	3
3	4
4	5
5	6

1. Calculate individual sums:

$$\sum x = 1 + 2 + 3 + 4 + 5 = 15$$

$$\sum y = 2 + 3 + 4 + 5 + 6 = 20$$

$$\sum x^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 1 + 4 + 9 + 16 + 25 = 55$$

$$\sum y^2 = 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 4 + 9 + 16 + 25 + 36 = 90$$

$$\sum xy = (1 \times 2) + (2 \times 3) + (3 \times 4) + (4 \times 5) + (5 \times 6) = 2 + 6 + 12 + 20 + 30 = 70$$

Example: Pearson's r

Calculate Pearson's r for variables x and y :

x	y
1	2
2	3
3	4
4	5
5	6

2. Plug values into Pearson's formula:

$$r = \frac{5 \times 70 - (15)(20)}{\sqrt{[5 \times 55 - (15)^2][5 \times 90 - (20)^2]}}$$

$$r = \frac{350 - 300}{\sqrt{[275 - 225][450 - 400]}}$$

$$r = \frac{50}{\sqrt{[50][50]}}$$

$$r = \frac{50}{\sqrt{2500}} = \frac{50}{50} = 1$$

Example: Pearson's r

Calculate Pearson's r for variables x and y :

x	y
1	2
2	3
3	4
4	5
5	6

3. Interpretation

- ▶ Pearson's $r = 1$
- ▶ Indicates a **perfect positive** linear relationship between the two variables
- ▶ As x increases, y increases in a perfectly linear manner

Spearman's ρ

Non-parametric correlation coefficient, monotonic relationship

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

Where:

- ▶ d_i is the difference between the ranks of each corresponding pair of values (x_i and y_i).
- ▶ n is the number of data points.
- ▶ The sum $\sum d_i^2$ is the sum of the squared differences of the ranks for each pair of values.

Example: Spearman's ρ

Calculate Spearman's ρ for the students' Math and Physics grades

Student	Math Grade	Physics Grade
Peter	1,0	1,7
John	3,0	3,0
Chris	2,3	1,3
Diana	3,7	3,3

1. Rank the values for both variables Math and Physics:

Student	Math Rank
Peter	1
John	3
Chris	2
Diana	4

Student	Physics Rank
Peter	2
John	3
Chris	1
Diana	4

Example: Spearman's ρ

Calculate Spearman's ρ for the students' Math and Physics grades

Student	Math Grade	Physics Grade
Peter	1,0	1,7
John	3,0	3,0
Chris	2,3	1,3
Diana	3,7	3,3

2. Calculate the difference in ranks d_i and square these differences:

Student	Math rank
Anja	1
Bernd	3
Christoph	2
Diana	4

d^2
1
0
1
0

Student	Physics rank
Anja	2
Bernd	3
Christoph	1
Diana	4

Example: Spearman's ρ

Calculate Spearman's ρ for the students' Math and Physics grades

Student	Math Grade	Physics Grade
Peter	1,0	1,7
John	3,0	3,0
Chris	2,3	1,3
Diana	3,7	3,3

3. Calculate ρ :

Student	Math rank
Anja	1
Bernd	3
Christoph	2
Diana	4

d^2
1
0
1
0

Student	Physics rank
Anja	2
Bernd	3
Christoph	1
Diana	4

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} = 1 - \frac{6 \cdot (1 + 0 + 1 + 0)}{4 \cdot (4^2 - 1)} = 1 - \frac{12}{60} = \mathbf{0.8}$$

Example: Spearman's ρ

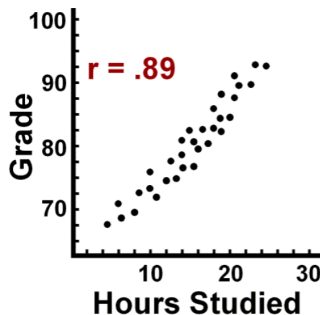
Calculate Spearman's ρ for the students' Math and Physics grades

Student	Math Grade	Physics Grade
Peter	1,0	1,7
John	3,0	3,0
Chris	2,3	1,3
Diana	3,7	3,3

4. Interpretation:

- ▶ $\rho = 0.8$ (close to 1) indicates a **strong positive monotonic relationship** between the two variables
- ▶ e.g., a strong tendency for the two variables to increase together

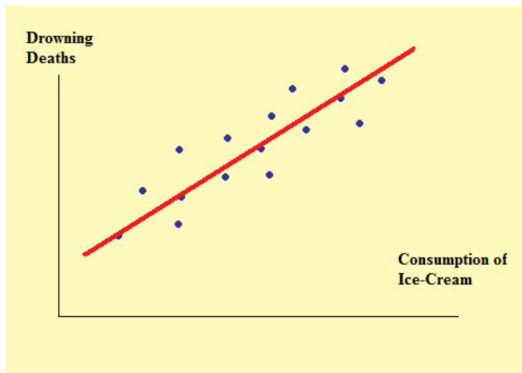
Correlation $\neq$ Causation



Correlation between two variables because:

- ▶ they are associated/there is a relationship between them
- ▶ they are both associated to a third factor
- ▶ by chance

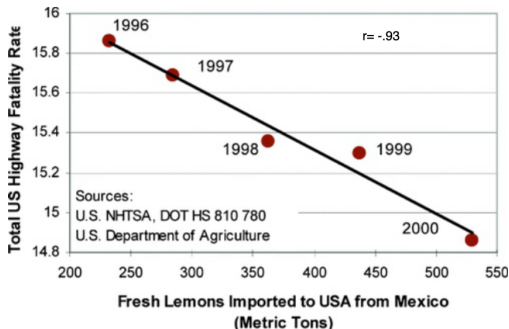
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Correlation $\neq$ Causation



Correlation between two variables because:

- ▶ they are associated/ there is a relationship between them
- ▶ they are both associated to a third factor
- ▶ by chance¹

¹<https://www.tylervigen.com/spurious-correlations>

Statistical Significance Test

Were the observed relationships in data (e.g., correlations) real or could they have occurred by chance?

- ▶ Formulate two contradicting hypotheses
 - ▶ Null Hypothesis H_0 : conservative, status quo, assumes no effect/independence
 - ▶ Alternative Hypothesis H_1 : assumes there is an effect/relationship
- ▶ choose a significance level (typically: $\alpha \in \{0.05, 0.01, 0.001\}$)
- ▶ perform a statistical test on empirical data and calculate p-value
- ▶ Based on evidence (compare p-value to significance level), reject one hypothesis and accept the other

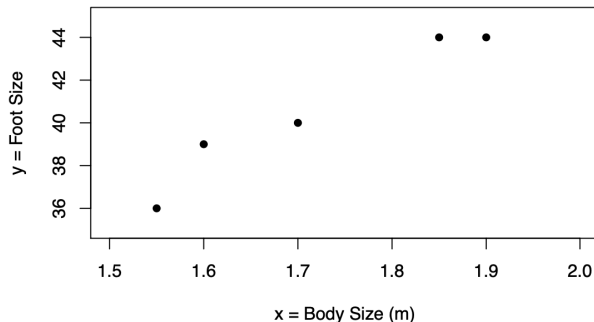
Statistical Significance & p-values

- ▶ **p-values:** quantifies the evidence against a *null hypothesis*
 - ▶ smaller p-values (< 0.05) suggest strong evidence against the null hypothesis, e.g., the relationship between variables is **statistically significant**
 - ▶ calculation depends on/specified for the exact test (test statistic and its distribution)
- ▶ **Confidence intervals:**
 - ▶ provides a range of values that likely includes the true correlation or regression coefficient, based on sample data
 - ▶ e.g., a 95% confidence interval means that if the study were repeated 100 times, 95% of the intervals would contain the true value
- ▶ **Combining the two:**
 - ▶ If a 95% confidence interval for a correlation includes 0, the relationship may not be statistically significant.
 - ▶ While if the interval does **not** include 0, the relationship is statistically significant at the 5% level
- ▶ **We do not go into detail on how to calculate the p-values or confidence intervals!**

Regression Basics

What is Regression?

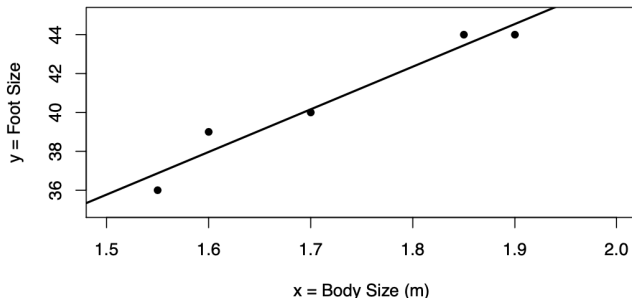
- ▶ Predicting a **continuous output** from one or more inputs
 - ▶ model and analyze the relationship between a **dependent variable** and one or more **independent variables**
 - ▶ aims to predict or understand how the dependent variable changes based on the predictors
 - ▶ Note: not a classification task (e.g., predicting discrete labels)



Simple Linear Regression

$$y = a + bx + \epsilon$$

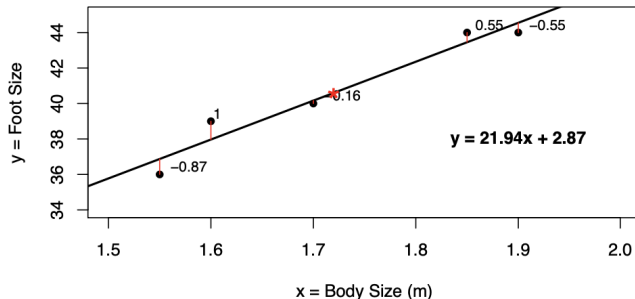
- ▶ Predict values for one attribute y (**predicted** attribute; dependent variable) given another attribute x (**predictor**; independent variable)
 - ▶ where, b is the slope parameter, a is the intercept, (and ϵ is some error)
- ▶ E.g., **predict** foot size (y) **by knowing** body size (x)



Ordinary Least Squares (OLS)

- ▶ Estimate the parameters a and b to get the best fit line
- ▶ **Objective:** Minimize sum of **squared residuals**, aka, the difference between the observed data points and the predicted values

$$\text{Minimize } \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - a - bx_i)^2$$



How can we initialize and train the regression model?

Initializing/Training a Linear Regression Model

Analytical Solution

For our simple linear regression model with one predictor, we can analytically derive the final formula $\hat{y} = a + bx$ using the correlation between x and y :

$$b = r \frac{s_y}{s_x}$$

$$a = \bar{y} - b\bar{x}$$

where

- ▶ r : Pearson's correlation coefficient between x and y
- ▶ $s_x = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$: Standard deviation of x for n datapoints
- ▶ $s_y = \sqrt{\frac{\sum (y - \bar{y})^2}{n-1}}$: Standard deviation of y
- ▶ $\bar{x} = \frac{\sum x}{n}$ mean of x

Initializing/Training a Linear Regression Model

Example

text length (in 100 words)	reading time (in minutes)
2	5
4	8
6	11
8	14
10	16

$$r = 0.997$$

$$s_x = 3.162$$

$$s_y = 4.743$$

Calculate Standard Deviations

x

$$\bar{x} = \frac{\sum x}{n} = \frac{2+4+6+8+10}{5} = 6$$

$$\begin{aligned}s_x &= \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}} \\&= \sqrt{\frac{\sum (x - 6)^2}{5 - 1}} \\&= \sqrt{\frac{(-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2}{4}} \\&= \sqrt{\frac{40}{4}} \\&= \sqrt{10} \\&= 3.162\end{aligned}$$

Calculate Standard Deviations

y

$$\bar{y} = \frac{\sum y}{n} = \frac{5+8+11+14+16}{5} = 11$$

$$\begin{aligned} s_y &= \sqrt{\frac{\sum (y - \bar{y})^2}{n - 1}} \\ &= \sqrt{\frac{\sum (y - 11)^2}{5 - 1}} \\ &= \sqrt{\frac{(-6)^2 + (-3)^2 + 0^2 + 3^2 + 6^2}{4}} \\ &= \sqrt{\frac{90}{4}} \\ &= \sqrt{22.5} \\ &= 4.743 \end{aligned}$$

Initializing/Training a Linear Regression Model

Example

text length (in 100 words)	reading time (in minutes)
2	5
4	8
6	11
8	14
10	16

$$r = 0.997, s_x = 3.162, s_y = 4.743$$

$$b = r \frac{s_y}{s_x} = 0.997 \cdot \frac{4.743}{3.162} = 1.496 \approx 1.5$$

$$a = \bar{y} - b\bar{x} = 10.8 - 1.5 \cdot 6 = 10.8 - 9 = 1.8$$

$$\hat{y} = 1.8 + 1.5x$$

Initializing/Training a Linear Regression Model

Example

text length (in 100 words)	reading time (in minutes)
2	5
4	8
6	11
8	14
10	16

$$\hat{y} = 1.8 + 1.5x$$

Predict the reading time for a text with 11,5 hundred words

$$\hat{y} = 1.8 + 1.5 \cdot 11.5 = 19.05$$

Multiple Linear Regression

Extending to Multiple Predictors

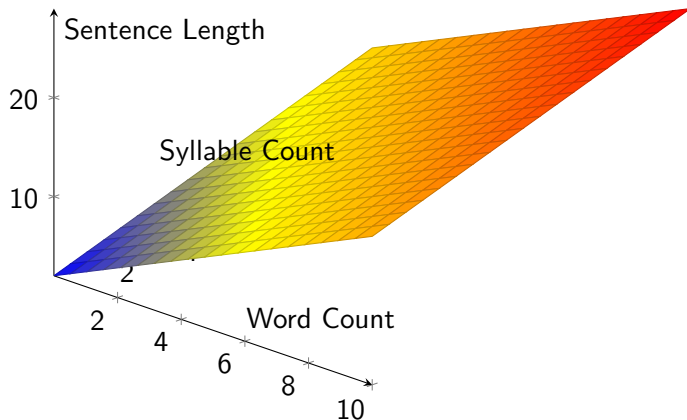
- ▶ Extend the concept of linear regression to include multiple predictors
- ▶ E.g., predict the **sentence length** based on two predictors:
 - ▶ **Word count** (x_1): the number of words in the sentence
 - ▶ **Syllable count** (x_2): the number of syllables in the sentence
- ▶ The multiple linear regression model is:

$$\text{Sentence Length} = a + b_1 \times \text{Word Count} + b_2 \times \text{Syllable Count} + \epsilon$$

where:

- ▶ a and b are the coefficients for word count and syllable count, respectively,
- ▶ c is the intercept,
- ▶ ϵ is the error term.

Sentence Length predicted by word and syllable count



example equation: $\hat{y} = 2 + 1.5x_1 + 1.2x_2$

Regression Evaluation & Diagnostics

Residual Analysis

Faraway (2005)

- **Residuals:** The differences between observed and predicted values
- Residual analysis checks if the assumptions of regression models are met, incl. constant variance (homoscedasticity) and linearity

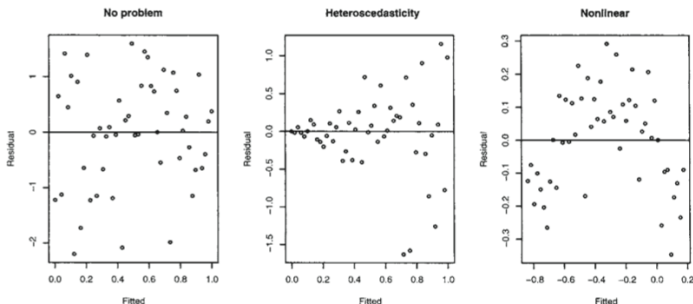


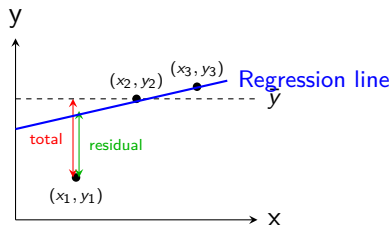
Figure 4.1 *Residuals vs. fitted plots—the first suggests no change to the current model while the second shows nonconstant variance and the third indicates some nonlinearity, which should prompt some change in the structural form of the model.*

Goodness of Fit: R-squared / Adjusted R-squared

Definition: Measures how well the regression model explains the variability in the data.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- ▶ y_i : Actual values
- ▶ $\hat{y}_i$: Predicted values
- ▶ $\bar{y}$: Mean of actual values



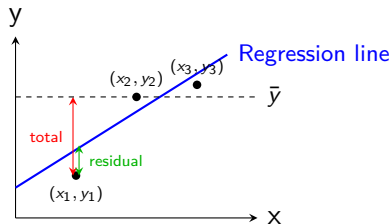
Adjusted R^2 : takes into account the number of predictors

Goodness of Fit: R-squared / Adjusted R-squared

Definition: Measures which proportion of the variance of y is explained by the model with the dependent variable x .

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- ▶ y_i : Actual values
- ▶ $\hat{y}_i$: Predicted values
- ▶ $\bar{y}$: Mean of actual values



Adjusted R^2 : takes into account the number of predictors

Goodness of Fit

- **Mean Squared Error (MSE)**: Average of the squared differences between predicted and actual values (residuals).

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- **Root Mean Squared Error (RMSE)**: Square root of MSE, providing the error in the same units as the data.

$$\text{RMSE} = \sqrt{\text{MSE}}$$

- **Mean Absolute Error (MAE)**: Average of the absolute errors, showing the typical error size (no penalty for large deviations).

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Calculate on train data to evaluate model fit, on test data to evaluate predictive performance

Practical Tools & Libraries

- ▶ Python (NumPy, SciPy, scikit-learn, statsmodels)
- ▶ R
- ▶ Example to be done in class on Tuesday

References I

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<https://online.stat.psu.edu/stat200/book>.